Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/10/2016
PI Name:	Donahue, Henry J. Ph.D.		
Project Title:	Integrated Regulation of Bone and Muscle Metabolism by Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Formerly at Pennsylvania State University, Hershey, PA, from 2012-2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):	Lang, Charles (The Pennsylvania State University)		
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Original project aims/objectives
Our long-term goal is to understand the structural and functional relationship between muscle atrophy (i.e., sarcopenia) and bone loss (i.e., osteopenia) which occurs as a result of extended exposure to the spaceflight environment. The specific objectives of this proposal are to: 1) Examine the time course for structural, functional, and molecular changes occurring in muscle and bone in response to mechanical unloading; 2) Examine the effect of inhibiting, or exaggerating, unloading-induced muscle loss on unloading-induced bone loss; 3) Examine the effect of inhibiting, or exaggerating, unloading-induced bone loss on unloading-induced muscle loss; and 4) Evaluate potential therapeutic strategies for prevention of muscle and bone loss during unloading. Upon completion of our specific aims, and based on our experimental approaches, we will recommend specific therapeutic strategies and countermeasures that target muscle and bone as an integrated unit and respect their inherent interdependency. To accomplish these objectives, we will use hindlimb suspension (HLS), a well established in vivo model of simulated microgravity unloading. Our overall

Task Description:	hypothesis is that unloading-induced osteopenia is at least partially dependent on unloading-induced sarcopenia and, conversely, unloading induced sarcopenia is at least partially dependent on unloading-induced osteopenia. Key findings 4-month-old male C57BL/6J mice subjected to hind limb suspension (HLS) for 3 weeks displayed a 7-9% decrease in lean mass vs. control from D7-21. Gastrocnemius (gastroc) mass decreased 8% by D7, and was maximally decreased 16% by D14 of HLS. Quadriceps (quad) lost a similar amount by D14. Control muscle mass did not change with age. Quantification of mRNA for the ubiquitin-E3 ligases Atrogin-1 and MuRF-1 revealed that Atrogin-1 mRNA levels in the gastroc and quad increased 99% and 122%, respectively, at D7 of HLS. Similar increases in Murf-1 mRNA levels were noted at D7. Both atrogenes returned to baseline by D14. Protein synthesis in gastroc and quad was reduced 30% from D7-14 of HLS, returning to baseline by D21. Cortical thickness and area fraction of the femur and tibia did not decrease until D14 of HLS. Cortical bone of controls did not change with age. HLS mice had lower distal femur trabecular bone volume/total volume (-22%) and trabecular thickness (-15%) by D14; however, the effects of HLS were often eliminated by D21 due to the decline of trabecular bone mass of controls. Femur strength was decreased approximately 13% by D14 of HLS. These results suggest that muscle atrophy precedes bone loss during unloading and may contribute to subsequent skeletal deficits. Sclerostin, encoded by the Sost gene, is a negative regulator of bone growth and Sost knockout (KO) mice are resistant to HLS-induced bone loss. 4-month-old male C57BL/6J mice globally deficient in Sost (Sost KO) and wild type (WT) controls were exposed to HLS or maintained in control cages for 2 weeks. Sost KO mice displayed greater bone volume relative to WT at baseline. Sost KO mice, relative to WT controls, displayed reduced loss of trabecular bone volume/total volume (-4% vs. -40%, respectively, $p<0.05$), trabecular thickness (-2% vs. -30% $p<0.05$) and trabecular number (+9% vs. -12% $p<0.05$) after HLS. HLS resulted in a similar loss of gastroc and quad muscle mass (14-18%) in Sost KO and WT mice. In a third study, we examined the hypothesis that exaggerating unloading-induced muscle loss via HLS + casting also exaggerates unloading-induced bone loss. Twenty 5-month-old male C57BL/6J mice were exposed to HLS with one limb unrestrained (HLS) and one limb casted (HLS+Cast), or maintained as ground control for 2 weeks. Trabecular bone volume/total volume decreased ($p<0.01$) in HLS and HLS+Cast limbs compared to controls, but there was no difference between HLS and HLS+Cast. There was no difference between any groups for trabecular number, thickness, or spacing. Gastroc and quad weight decreased 10% in HLS limbs and 20% in HLS+Cast compared to GC ($p<0.05$). Protein synthesis was decreased in HLS and HLS+Cast compared to GC ($p<0.05$); however, HLS+Cast had a 15% increase in protein synthesis compared to HLS ($p<0.05$). Measurements of mTOR signaling (S6K1 and 4E-BP1 phosphorylation) correlated with protein synthesis measurements. There was no difference in Atrogin-1 or MuRF-1 mRNA between any groups. There was, however, an increase in autophagy associated proteins LC3-II/I, Atg7, and Atg5-12 in the HLS+Cast limb compared to both HLS and GC ($p<0.05$). Impact Our results suggest that muscle atrophy precedes bone loss during unloading and may contribute to subsequent skeletal deficits. Furthermore, protecting bone from HLS-induced bone loss via Sost deficiency did not protect against HLS-induced muscle. Additionally, casting exacerbates unloading-induced muscle loss via activation of the autophagy degradation pathway. Surprisingly, casting did not cause an exacerbation of bone loss. Taken together, our results suggest that HLS-induced muscle loss occurs before HLS-induced bone loss and that sarcopenia is not necessarily coupled to osteopenia. These findings should be considered when developing countermeasures for microgravity induced bone and muscle loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	On the ground, concomitant bone and muscle loss are frequently experienced by individuals confined to bed for long durations, whether a result of neurological injury, trauma, or neuromuscular disease (e.g., cerebral palsy, muscular dystrophy). In these patients, and in healthy patients subjected to extended bed rest, a significant loss of muscle and bone occurs. In the case of recoverable disease, or when ambulation is still possible, this leaves patients at risk for falls and catastrophic fracture. Furthermore, age-related bone and muscle loss is a serious problem in our increasingly aged population. Understanding the interdependence of muscle and bone loss during spaceflight, and the development of countermeasure strategies, will provide a framework for the development of preventative treatments for astronauts and the clinical population, especially those suffering from osteopenia and sarcopenia.
Task Progress:	We have completed aims 1 and the first part of aim 3 and aim 2. Results from aim 1 have been published; results of aim 3 have been submitted and the manuscript is under revision and results of aim 2 are being analyzed. Aim 4 was not addresed as NSBRI (National Space Biomedical Research Institute) staff felt it would be over ambitious for this project.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
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